TEXORA · LECTURE SHEET

Fabric Geometry

Peirce's geometrical model of the plain weave: thread spacing, crimp and crimp height, the relation $h_1 + h_2 = d_1 + d_2$, fractional cover and cover factor, and the jammed limit that tells a designer how close a cloth can possibly be set.

FM-4203 · FABRIC MECHANICS · TOPIC 3 · FABRIC MANUFACTURING SPECIALIZATION

1. The variables of a woven cloth

A woven fabric is specified geometrically by a short list of measurable quantities, for each thread system (suffix 1 = warp, 2 = weft):

- **Threads per unit length** N (ends/cm, picks/cm) and its reciprocal, the **thread spacing** p = 1 ÷ N.
- **Yarn diameter** d (Topic 2).
- **Crimp** c = (l − p) ÷ p, where l is the length of yarn in one repeat of the cloth measured along the yarn, p the repeat length measured along the cloth. Crimp is the fabric-level obliquity tax: it is the extra yarn the structure consumes, and the same number appears in production as *take-up*.
- **Crimp height (amplitude)** h — the peak-to-peak excursion of the yarn's centre line across the cloth thickness.
- **Fabric thickness** t, measured under a stated pressure.

REFERENCE CLOTH — USED THROUGHOUT THIS SHEET

Plain weave, warp 20 tex at 28 ends/cm, weft 24 tex at 24 picks/cm, diameters from Topic 2: d_1 = 0.167 mm, d_2 = 0.183 mm.
Spacings: p_1 = 10 ÷ 28 = **0.357 mm** (between adjacent ends); p_2 = 10 ÷ 24 = **0.417 mm** (between adjacent picks).
Combined diameter D = $d_1 + d_2$ = 0.167 + 0.183 = **0.350 mm** — the distance between the two yarn centre lines wherever warp and weft are in contact.

2. Peirce's model

F. T. Peirce's 1937 model — still the frame of all fabric geometry — idealises the plain weave as follows: yarns are **circular in cross-section and inextensible**; between crossovers each yarn runs

straight; at a crossover it wraps the crossing yarn; and the two centre lines stay exactly D apart wherever they touch. One exact consequence organises everything:

$$h_1 + h_2 = d_1 + d_2 = D$$

The crimp heights are not independent: whatever wave the warp does not take, the weft must. A rigid, tightly sized warp stays nearly straight ($h_1 \to 0$) and throws all the crimp onto the weft ($h_2 \to D$); a slack warp in a dense weft takes the wave itself. Finishing that relaxes the cloth lets the two systems renegotiate their shares — which is why dimensions change in washing even though no yarn length changes (the *l* values are fixed; the h values and spacings move).

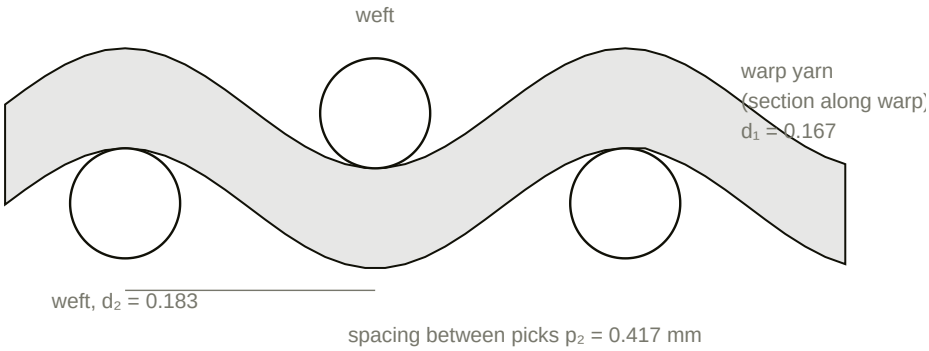


Figure 1 — Peirce cross-section of the reference cloth, drawn to scale (300 px/mm). Warp d_1 = 0.167 mm taking crimp height h_1 = 0.20 mm; weft d_2 = 0.183 mm taking h_2 = 0.15 mm; $h_1 + h_2$ = 0.35 mm = D, and at every crossover the centre lines are exactly D apart — the warp band just touches each weft circle.

From the geometry Peirce derived the working relations between crimp, spacing and angle (each yarn leaves a crossover at an angle θ to the cloth plane, and l, p, h and θ are bound together: roughly, higher crimp at fixed spacing means larger θ and larger h). For the practising designer the model's three deliverables are the ones of Sections 3–5: **cover**, the **jammed limit**, and a **thickness estimate** — at a crossover the cloth is at least D thick, and overall thickness lies between D and D + the smaller diameter depending on how the crimp is shared.

3. Fractional cover and cover factor

The fraction of the cloth area hidden by one thread system is its **fractional cover**: C = N × d (with N per cm and d in cm). The two systems overlap at crossovers, so combined cover is $C_1 + C_2 - C_1C_2$, not the sum.

WORKED EXAMPLE — COVER OF THE REFERENCE CLOTH

Warp: $C_1 = 28 \times 0.0167 =$ **0.47**. Weft: $C_2 = 24 \times 0.0183 =$ **0.44**.

Combined cover = 0.47 + 0.44 − (0.47 × 0.44) = 0.91 − 0.21 = **0.70**.

So about seven-tenths of the area is covered by yarn and three-tenths is open interstice — a moderately open plain cloth, as its air permeability would confirm. In **cover factor** form (metric convention K = threads/cm × √tex ÷ 10): warp $K_1 = 28 \times \sqrt{20} \div 10 =$ **12.5**; weft $K_2 = 24 \times \sqrt{24} \div 10 =$ **11.8** — the ordinary commercial range for cotton plain weaves, and the numbers a designer quotes when re-engineering the cloth in a different count at the same closeness.

4. The jammed limit: how close can a cloth be set?

Geometry also sets the limit of design. In a plain weave an incoming pick must pass over and under alternate ends; as the spacing shrinks toward the yarn diameter the shed can no longer form and weaving becomes impossible. The absolute ceiling for either system, from $p \geq d$, is:

$$N_{max} = 1 \div d \quad \text{(per unit length)}$$

WORKED EXAMPLE — MAXIMUM SETT

For a yarn of d = 0.20 mm: $N_{max} = 10 \div 0.20 =$ **50 threads/cm** — and real weaving stops well short of it, because the crossing system also needs room to displace (Peirce's full treatment gives lower practical limits for balanced cloths, and a warp can be set closer than the limit only by throwing nearly all crimp onto the weft, $h_2 \rightarrow D$). The lesson the limit teaches is comparative: of two cloths, the one nearer its jammed state is the firmer, less extensible, harder to weave — a statement about ratios, available before any loom is threaded.

5. Extensions of the model

Three refinements matter beyond plain weave. **Flattened sections:** in real cloth the yarn at crossovers flattens toward an ellipse (major axis up to ~1.3–1.5× the minor for cotton), which is why measured thickness is usually less than D and real cover greater than the circular model predicts — the model's errors are systematic, hence correctable. **Other weaves:** floats let yarns stay straighter (lower h, lower crimp) and pack by nesting side by side; the same variables describe them, with the repeat and float length entering the crimp relations. **Biaxial mechanics:** the geometry explains the shape of the fabric load–extension curve — an initial easy extension while crimp is exchanged between the systems (warp straightens, weft crimps more: *crimp interchange*), stiffening sharply as the loaded system approaches straightness and the yarn's own modulus takes over. Topic 4 picks the story up from exactly this point.

Glossary

Thread spacing p — centre-to-centre distance of adjacent parallel threads, 1/N. **Crimp c** — (yarn length − cloth length)/cloth length per repeat. **Crimp height h** — peak-to-peak excursion of a yarn

centre line. **Fractional cover** — N × d for one system; combined $C_1 + C_2 - C_1C_2$. **Cover factor** — count-normalised closeness, K = threads/cm × √tex/10 (metric). **Jammed state** — closest possible setting, spacing approaching yarn diameter. **Crimp interchange** — transfer of crimp between warp and weft under load or relaxation.

Key points

• Peirce: circular, inextensible yarns, centre lines $D = d_1 + d_2$ apart at contact, and $h_1 + h_2 = D$ — crimp is shared, and finishing renegotiates the shares.
• Reference cloth: $p_1 = 0.357$ mm, $p_2 = 0.417$ mm, D = 0.350 mm; cover 0.47/0.44, combined 0.70; cover factors 12.5 and 11.8.
• Jammed ceiling $N_{max} = 1/d$ (50/cm at d = 0.20 mm); practical cloths stop short of it, and closeness *relative* to the limit is the firmness variable.
• Real yarns flatten at crossovers; floats and crimp interchange extend the same geometry to other weaves and to the shape of the load–extension curve.

Practice

1. A weft of diameter 0.25 mm is inserted at 20 picks/cm. Find its fractional cover, and the combined cover with a warp of fractional cover 0.56.
2. What is the maximum sett (threads/cm) for a yarn of diameter 0.167 mm, and why can a real plain weave not be woven at that sett in both systems at once?
3. A fabric's warp crimp is measured at 9% and its weft crimp at 3%. After a relax-and-wash finish the warp crimp falls to 6%. State, with the governing relation, what must have happened to the weft crimp and to the fabric's length and width.
4. Explain crimp interchange in the first part of a warp-way tensile test.

Answer key

1. $C_2 = 20 \times 0.025 =$ **0.50**. Combined = 0.56 + 0.50 − 0.56 × 0.50 = 1.06 − 0.28 = **0.78**.
2. $N_{max} = 10 \div 0.167 \approx$ **60 threads/cm**. At that spacing in both systems simultaneously there is no room for either yarn to displace across the other ($h_1 + h_2 = D$ cannot be satisfied with both systems at p = d): the shed cannot form; weaving needs spacing reserve in at least one system.
3. Yarn lengths l are unchanged by washing, and $h_1 + h_2 = D$ must still hold with a smaller warp wave: the weft crimp height and crimp must have **risen** (crimp interchange). A falling warp crimp means the warp straightened, so the fabric **lengthened warp-way or the weft took up the difference by contracting the width** — in practice the cloth relaxes to a new balance with changed dimensions in

both directions, the warp-way dimension increasing at the expense of crimp, the weft-way adjusting oppositely.

4. Initially the warp yarns are crimped and the test mostly *straightens* them: their h_1 falls, and because $h_1 + h_2 = D$, the weft's crimp rises by the same amount — extension is bought by geometry, not by stretching yarn, so the modulus is low. Once the warp is nearly straight, further extension must stretch the yarn itself and the curve steepens to the yarn modulus.
